

CBSE Class 12 Chemistry — The d- and f-Block Elements

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQs / Assertion-Reason / Very Short Answer)

- Q1.** Out of the following transition metals, the maximum number of oxidation states are shown by: (a) Chromium ($Z = 24$) (b) Manganese ($Z = 25$) (c) Iron ($Z = 26$) (d) Titanium ($Z = 22$) [PYQ 2020] [Confidence: Very High]
- Q2.** The intense purple colour of $KMnO_4$ is due to: (a) $d - d$ transitions (b) Charge transfer from ligand to metal (c) Unpaired electrons in the d orbital of Mn (d) Charge transfer from metal to ligand [PYQ SQP 2022] [Confidence: Very High]
- Q3.** Which of the following ions is colourless in an aqueous solution? (a) Ti^{4+} (b) Cr^{3+} (c) V^{3+} (d) Fe^{2+} [PYQ SQP Term-2 2022] [Confidence: Very High]
- Q4.** The name 'inner transition metals' is often used to refer to: (a) Lanthanoids (b) Actinoids (c) Both lanthanoids and actinoids (d) d-block elements [PYQ SQP 2025-26] [Confidence: High]
- Q5.** Which of the following is not considered a true transition element? (a) Scandium (Sc) (b) Iron (Fe) (c) Zinc (Zn) (d) Copper (Cu) [Delhi Gov. SQP Term-2 2022] [Confidence: Very High]
- Q6.** The calculated spin-only magnetic moment for a divalent ion in an aqueous solution if its atomic number is 25 is: (a) 5.92 BM (b) 4.90 BM (c) 3.87 BM (d) 1.73 BM [Practice] [Confidence: High]
- Q7.** Which of the following exhibits a diamagnetic behaviour? (a) Sc^{3+} (b) Cr^{3+} (c) Mn^{2+} (d) Fe^{3+} [PYQ 2020] [Confidence: High]
- Q8.** In the 3d series, which element has the lowest enthalpy of atomisation? (a) Sc (b) Cr (c) Cu (d) Zn [Practice] [Confidence: High]
- Q9.** Eu^{2+} acts as a strong reducing agent because it tends to attain the highly stable electronic configuration of: (a) Eu^{4+} (b) Eu^{3+} (c) Eu^{+} (d) Eu [PYQ 2019] [Confidence: Medium]
- Q10.** The highest oxidation state of transition metals is usually exhibited in their: (a) Chlorides and Bromides (b) Oxides and Fluorides (c) Hydrides and Carbides (d) Sulphides and Phosphides [PYQ SQP Term-2 2022] [Confidence: Very High]
- Directions for Q11 to Q15:** A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

- Q11.** Assertion (A): Transition metals have high melting points. Reason (R): Transition metals have completely filled d-orbitals. [PYQ 2020] [Confidence: Very High]
- Q12.** Assertion (A): Magnetic moment values of actinides are lesser than the theoretically predicted values. Reason (R): Actinide elements are strongly paramagnetic. [PYQ SQP 2022] [Confidence: High]
- Q13.** Assertion (A): Highest oxidation state is exhibited by transition metal lying in the middle of the series. Reason (R): The highest oxidation state exhibited corresponds to the sum of $(n - 1)d$ and ns electrons. [CBSE PDF] [Confidence: High]
- Q14.** Assertion (A): Fe^{3+} is more stable than Fe^{2+} . Reason (R): Fe^{3+} has a stable $3d^5$ half-filled configuration while Fe^{2+} has a $3d^6$ configuration. [CBSE PDF] [Confidence: Very High]
- Q15.** Assertion (A): Group 12 elements (Zn, Cd, Hg) are not considered as transition metals. Reason (R): Transition metals are those which have an incompletely filled d-shell in their ground state or in any of their common oxidation states. [CBSE PDF] [Confidence: Very High]
- Q16.** Write the formula of an oxo-anion of manganese (Mn) in which it shows the oxidation state equal to its group number. [PYQ 2017] [Confidence: High]
- Q17.** Identify the ion which is a strong reducing agent out of Cr^{2+} and Mn^{3+} despite both having a d^4 configuration. [PYQ 2020] [Confidence: Very High]
- Q18.** Complete the following chemical equation: $3MnO_4^{2-} + 4H^+ \rightarrow$ [PYQ 2015] [Confidence: High]
- Q19.** Write the electronic configuration of Lu^{3+} (At. No. = 71). [CBSE PDF] [Confidence: Medium]
- Q20.** Out of the following, which one would feel attraction when placed in a magnetic field: $Co^{2+}, Ag^+, Ti^{4+}, Zn^{2+}$? [PYQ SQP Term-2 2022] [Confidence: High]
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2-Mark Questions (Short Answer)

- Q21.** Are the enthalpies of atomisation of Zinc and Copper matched correctly in the following table? Justify your answer. Element: Zinc | Enthalpy: 339 kJ/mol Element: Copper | Enthalpy: 130 kJ/mol [PYQ SQP 2025-26] [Confidence: Very High]
- Q22.** Account for the following: Mn^{2+} is much more resistant than Fe^{2+} towards oxidation. [PYQ SQP Term-2 2022] [Confidence: Very High]
- Q23.** Write any two consequences of Lanthanoid contraction. [PYQ Term-2 2022] [Confidence: Very High]
- Q24.** Explain why transition metals and their compounds are well-known for their catalytic activity. [Delhi Gov. SQP Term-2 2022] [Confidence: High]
- Q25.** Account for the following: Zinc is a comparatively soft metal, whereas iron and chromium are typically hard. [PYQ SQP Term-2 2022] [Confidence: High]

- Q26.** Give two reasons why transition metals frequently form interstitial compounds. What is one physical property of such compounds? [PYQ 2020 / Practice] [Confidence: High]
- Q27.** It has been observed that the first ionization energy of the 5d series of transition elements is higher than that of the 3d and 4d series. Explain why. [PYQ SQP Term-2 2022] [Confidence: Medium]
- Q28.** Calculate the spin-only magnetic moment of Co^{2+} ($Z = 27$) by writing its electronic configuration. [PYQ 2020] [Confidence: Very High]
- Q29.** Account for the following: Copper (I) compounds are generally white/colourless whereas copper (II) compounds are coloured. [PYQ 2020] [Confidence: Very High]
- Q30.** Out of sulphuric acid and hydrochloric acid, which acid will you prefer for permanganate titrations and why? [PYQ SQP 2025-26] [Confidence: Very High]
- Q31.** Account for the fact that Ce^{4+} is a strong oxidising agent. [PYQ 2023] [Confidence: High]
- Q32.** Why do transition metals form alloys easily? Give an example. [PYQ 2023 / Practice] [Confidence: High]

3-Mark Questions (Short Answer)

- Q33.** Account for the following observations: (i) E° value for the $\text{Mn}^{3+}/\text{Mn}^{2+}$ couple is much more positive than that for $\text{Cr}^{3+}/\text{Cr}^{2+}$. (ii) Sc^{3+} is colourless whereas Ti^{3+} is coloured in an aqueous solution. (iii) Actinoids show a wide range of oxidation states. [PYQ 2023] [Confidence: Very High]
- Q34.** Complete and balance the following chemical equations: (i) $\text{MnO}_4^- + \text{I}^- + \text{H}^+ \rightarrow$ (ii) $\text{Cr}_2\text{O}_7^{2-} + \text{Sn}^{2+} + \text{H}^+ \rightarrow$ (iii) $2\text{MnO}_2 + 4\text{KOH} + \text{O}_2 \xrightarrow{\Delta}$ [PYQ 2019/2016] [Confidence: Very High]
- Q35.** Answer the following questions based on d-block elements: (i) Why does Mn show the highest oxidation state of +7 with oxygen but only +4 with fluorine? (ii) Why is Cu^+ unstable in aqueous solution? (iii) In transition elements, ions of the same charge show a progressive decrease in radius with increasing atomic number. Why? [PYQ 2016/2020/2022] [Confidence: High]
- Q36.** Which of the following elements will: (i) exhibit similar magnetic behaviour and why? Magnesium ($Z = 12$), Chromium ($Z = 24$), Iron ($Z = 26$) and Molybdenum ($Z = 42$). (ii) form white salts and why? Zinc ($Z = 30$), Scandium ($Z = 21$), Nickel ($Z = 28$) and Vanadium ($Z = 23$). [PYQ SQP 2025-26] [Confidence: High]
- Q37.** When chromite ore FeCr_2O_4 is fused with NaOH in the presence of air, a yellow-coloured compound (X) is obtained which on acidification with dilute sulphuric acid gives a compound (Y). Compound (Y) on reaction with KCl forms an orange-coloured crystalline compound (Z). (i) Write the formulae of the compounds (X), (Y) and (Z). (ii) Write one use of compound (Z). [PYQ 2016] [Confidence: Very High]

Q38. Account for the following: (i) Zirconium (**Zr**) and Hafnium (**Hf**) have almost similar atomic radii. (ii) The highest oxidation states of transition metals are exclusively shown in their oxides and fluorides. (iii) Copper does not displace hydrogen from dilute acids. [PYQ 2015/2022] [Confidence: High]

Q39. Complete the following equations: (i) $5\text{NO}_2^- + 2\text{MnO}_4^- + 6\text{H}^+ \rightarrow$ (ii) $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{I}^- \rightarrow$ (iii) $\text{KMnO}_4 \xrightarrow{\Delta}$ [PYQ SQP 2025-26 / 2016 / 2017] [Confidence: Very High]

Q40. Give reasons: (i) The $E^\circ(\text{V}^{3+}/\text{V}^{2+})$ value for vanadium is comparatively low (negative). (ii) Transition metals have high enthalpies of atomisation. (iii) The atomic radius of **d**-block elements slightly increases at the end of each period. [PYQ SQP 2022 / Practice] [Confidence: Medium]

Q41. Answer the following: (i) Why are all copper halides known except copper iodide? (ii) What is the effect of pH on a dichromate ion solution? (iii) Name an element of the 3d series that does not show variable oxidation states. [PYQ SQP 2022 / 2015] [Confidence: High]

Q42. Complete and balance the following chemical equations: (i) $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow$ (ii) $\text{Cr}_2\text{O}_7^{2-} + \text{Fe}^{2+} + \text{H}^+ \rightarrow$ (iii) $\text{MnO}_4^- + \text{S}_2\text{O}_3^{2-} + \text{H}_2\text{O} \rightarrow$ (in neutral/faintly alkaline medium) [PYQ 2017/2018/2016] [Confidence: Very High]

5-Mark Questions (Long Answer / Case-Study)

Q43. (A) Describe the preparation of Potassium permanganate (**KMnO₄**) from pyrolusite ore (**MnO₂**). Write the balanced chemical equations for the two major steps involved. (B) Complete the following ionic equation: $\text{Cr}_2\text{O}_7^{2-} + 2\text{OH}^- \rightarrow$ (C) Write one similarity and one difference between the chemistry of Lanthanoids and Actinoids. [PYQ 2023] [Confidence: Very High]

Q44. When **MnO₂** is fused with **KOH** in the presence of **KNO₃** as an oxidizing agent, it gives a dark green compound (A). Compound (A) disproportionates in acidic solution to give a purple compound (B). An alkaline solution of compound (B) oxidises **KI** to compound (C) whereas an acidified solution of compound (B) oxidises **KI** to (D). (i) Identify (A), (B), (C), and (D). (ii) Write the balanced chemical equations for the formation of (C) and (D). [PYQ 2019] [Confidence: High]

Q45. Account for the following: (i) **Ti(IV)** is more stable than **Ti(II)** or **Ti(III)**. (ii) Halides of transition metals are more stable in higher oxidation states than in lower ones. (iii) Enthalpy of atomisation of Zinc is the lowest in the 3d series. (iv) **Sc³⁺** is diamagnetic but **Cu²⁺** is paramagnetic. (v) **Cr²⁺** is a stronger reducing agent than **Fe²⁺**. [PYQ SQP 2022 / Practice] [Confidence: Very High]

Q46. (A) Describe the preparation of Potassium dichromate (**K₂Cr₂O₇**) from chromite ore. Write the balanced chemical equations. (B) Calculate the magnetic moment of **Fe²⁺** and **Mn²⁺**. Which is more paramagnetic? [Practice based on PYQ Themes] [Confidence: High]

Q47. Case Study: Lanthanoid and Actinoid Contraction The decrease in atomic and ionic radii across the lanthanoid series is known as the lanthanoid contraction. A similar phenomenon occurs in the actinoids. This contraction is due to the imperfect shielding of one f-electron by another in the same subshell. (i) What is the main chemical consequence of the lanthanoid contraction regarding the 4d and 5d series? (1 Mark) (ii) Why is actinoid contraction greater from element to element than lanthanoid contraction? (1 Mark) (iii) Write the electronic configuration of Cerium (Ce, $Z = 58$) and explain why Ce^{4+} is a good analytical oxidizing agent. (3 Marks) [Practice based on CBSE PDF] [Confidence: Very High]

Q48. (A) Arrange the following monoxides of transition metals on the basis of decreasing basic character: TiO , VO , CrO , FeO . Give a reason. (B) Explain why $CuSO_4 \cdot 5H_2O$ is blue while $ZnSO_4$ and anhydrous $CuSO_4$ are colourless. (C) Write the formula of the oxo-anion of Chromium (Cr) in which it shows an oxidation state equal to its group number. [CBSE PDF / PYQ 2017] [Confidence: High]

Q49. (A) Transition metals act as excellent catalysts. Give two specific reasons for this property and provide one industrial example. (B) Why do transition elements form a large number of complex compounds? (Provide two reasons). (C) What is the effect of increasing pH on a solution of potassium dichromate? Support your answer with a chemical equation. [Practice / PYQ 2017] [Confidence: High]

Q50. Case Study: Properties of Transition Metals The d-block elements form alloys easily because of similar atomic sizes. They also form interstitial compounds where small atoms like H, C, N are trapped inside the crystal lattice. (i) Why do transition metals form alloys easily? (1 Mark) (ii) Why are interstitial compounds chemically inert and hard? (1 Mark) (iii) The lowest oxide of a transition metal is basic, whereas the highest is amphoteric or acidic. Give one example of a basic oxide and one of an acidic oxide of Manganese. (2 Marks) (iv) Name the transition metal of the 3d series which does not exhibit variable oxidation states. (1 Mark) [Practice based on PYQ themes] [Confidence: Very High]

Answer Key

Q1. (b) Manganese ($Z = 25$). Shows oxidation states from +2 to +7. **Q2.** (b) Charge transfer from ligand (oxygen) to metal (manganese). **Q3.** (a) Ti^{4+} . It has a d^0 electronic configuration (no unpaired electrons for d-d transition). **Q4.** (c) Both lanthanoids and actinoids. **Q5.** (c) Zinc (Zn). It has a completely filled d-subshell ($3d^{10}$) in its ground state and common oxidation state. **Q6.** (a) 5.92 BM. A divalent ion (M^{2+}) of $Z=25$ is Mn^{2+} ($3d^5$). $n = 5$. $\mu = \sqrt{5(5+2)} = \sqrt{35} \approx 5.92$ BM. **Q7.** (a) Sc^{3+} . It has a $3d^0$ configuration (no unpaired electrons). **Q8.** (d) Zn . Due to the absence of unpaired d-electrons, interatomic metallic bonding is the weakest. **Q9.** (b) Eu^{3+} . (Eu is $[Xe]4f^76s^2$. Eu^{2+} is $4f^7$, but it easily loses one more electron to reach the common +3 state, acting as a reducing agent). **Q10.** (b) Oxides and Fluorides. Oxygen and fluorine are highly electronegative and small, able to stabilize high oxidation states. **Q11.** (c) A is true but R is false. (High melting points are due to

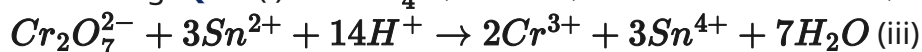
strong metallic bonding resulting from unpaired d-electrons, not completely filled d-orbitals).

Q12. (b) Both A and R are true but R is not the correct explanation of A. (Lesser moments are due to strong spin-orbit coupling and quenching of orbital angular moment, not just paramagnetism). **Q13.** (a) Both A and R are true and R is the correct explanation of A. (e.g., Mn shows +7 using $3d^5 4s^2$). **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (a) Both A and R are true and R is the correct explanation of A. **Q16.** Permanganate ion, MnO_4^- . (Mn group number is 7, ox state in MnO_4^- is +7). **Q17.** Cr^{2+} . (Cr^{2+} oxidizes to Cr^{3+} , gaining a stable t_{2g}^3 half-filled configuration. Mn^{3+} reduces to Mn^{2+} , gaining a stable $3d^5$ half-filled configuration). **Q18.** $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$ **Q19.** $[Xe]4f^{14}$. **Q20.** Co^{2+} . (It has $3d^7$ configuration with 3 unpaired electrons, so it is paramagnetic and feels attraction. Others are diamagnetic: $Ag^+(4d^{10})$, $Ti^{4+}(3d^0)$, $Zn^{2+}(3d^{10})$).

Q21. No. The values are reversed. Enthalpy of atomisation of Zn is lower (130 kJ/mol) because it has a completely filled d-subshell ($3d^{10}$) resulting in weak metallic bonding, whereas Cu has stronger interatomic interactions (339 kJ/mol). **Q22.** Mn^{2+} has a highly stable exactly half-filled $3d^5$ electronic configuration, so it resists losing another electron. $Fe^{2+}(3d^6)$ easily loses one electron to form the stable $Fe^{3+}(3d^5)$ configuration. **Q23.** (1) Radii of members of the 4d and 5d series are almost identical (e.g., Zr and Hf). (2) Separation of lanthanoids becomes difficult due to identical chemical properties resulting from similar sizes. **Q24.** Transition metals exhibit variable oxidation states and can provide a large surface area for adsorption of reactant molecules, lowering the activation energy of the reaction. **Q25.** Zn has a completely filled d-subshell ($3d^{10}$), preventing d-electron participation in metallic bonding (weak metallic bonds = soft). Fe and Cr have unpaired d-electrons, allowing strong interatomic metallic bonding (hard). **Q26.** They form interstitial compounds because small non-metal atoms (H, C, N) easily fit into the empty spaces (interstices) of their metallic lattices. Property: They are very hard and have higher melting points than the pure metals. **Q27.** In the 5d series, the intervening 4f electrons provide poor shielding of the nucleus. This results in a higher effective nuclear charge acting on the outer electrons, thereby increasing the ionization energy. **Q28.** Co is $[Ar]3d^7 4s^2$. Co^{2+} is $[Ar]3d^7$. It has 3 unpaired electrons ($n = 3$). $\mu = \sqrt{3(3+2)} = \sqrt{15} \approx 3.87 \text{ BM}$. **Q29.** Cu^+ has a $3d^{10}$ configuration (no unpaired electrons, no d-d transition = colourless). Cu^{2+} has a $3d^9$ configuration (one unpaired electron, d-d transition possible = coloured). **Q30.** Sulphuric acid is preferred. Hydrochloric acid cannot be used because $KMnO_4$ will oxidise HCl to chlorine gas (Cl_2), interfering with the titration. **Q31.** Cerium ($Z = 58$) easily loses its 4 electrons to attain the stable noble gas configuration of Xenon (Ce^{4+}). However, the most stable state for lanthanoids is +3. So, Ce^{4+} readily gains an electron to form Ce^{3+} , acting as a strong oxidising agent. **Q32.** Because of similar atomic sizes (differing by less than 15%), atoms of one transition metal can easily take the positions of another in the crystal lattice. Example: Brass (Cu and Zn) or Bronze.

Q33. (i) Mn^{3+} to Mn^{2+} results in a stable half-filled $3d^5$ configuration, driving the reduction (positive E°). Cr^{3+} to Cr^{2+} is less favorable as $Cr^{3+}(t_{2g}^3)$ is already highly stable in aqueous solution. (ii) Sc^{3+} is $3d^0$ (no unpaired electrons for d-d transition = colourless). Ti^{3+}

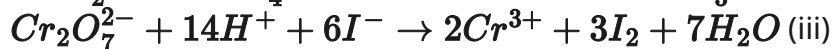
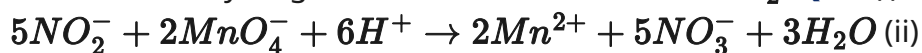
is $3d^1$ (unpaired electron allows d-d transition = coloured). (iii) The 5f, 6d, and 7s orbitals in actinoids have comparable energies, allowing electrons from all these subshells to participate in bonding. **Q34.** (i) $2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{2+} + 5I_2 + 8H_2O$ (ii)



$2MnO_2 + 4KOH + O_2 \xrightarrow{\Delta} 2K_2MnO_4 + 2H_2O$ **Q35.** (i) Oxygen can form multiple (pi) bonds with Mn, stabilizing the +7 state, whereas fluorine can only form single bonds. (ii) Cu^+ disproportionates in water: $2Cu_{(aq)}^+ \rightarrow Cu_{(aq)}^{2+} + Cu_{(s)}$, driven by the high hydration enthalpy of Cu^{2+} . (iii) Due to poor shielding by added d-electrons, the effective nuclear charge increases progressively, pulling the electron cloud closer and decreasing the radius.

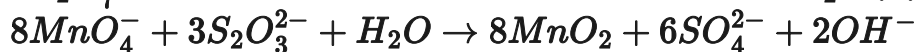
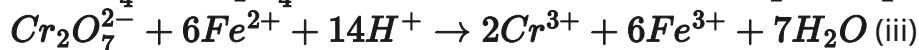
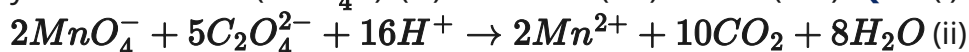
Q36. (i) Chromium and Molybdenum. Cr ($3d^5 4s^1$) and Mo ($4d^5 5s^1$) both have 6 unpaired electrons, thus similar magnetic moments. (ii) Zinc and Scandium. Zn^{2+} is $3d^{10}$ and Sc^{3+} is $3d^0$. Since neither has unpaired d-electrons, they cannot undergo d-d transitions and form white/colourless salts.

Q37. (i) X = Na_2CrO_4 (Sodium chromate), Y = $Na_2Cr_2O_7$ (Sodium dichromate), Z = $K_2Cr_2O_7$ (Potassium dichromate). (ii) Z ($K_2Cr_2O_7$) is widely used as a strong oxidising agent in volumetric analysis and organic synthesis. **Q38.** (i) Due to the lanthanoid contraction, the filling of 4f orbitals leads to poor shielding, causing 5d elements (Hf) to be similar in size to 4d elements (Zr). (ii) Oxygen and fluorine are highly electronegative and small in size, enabling them to bring out the highest oxidation states. (iii) Copper has a positive standard reduction potential ($E^\circ = +0.34 V$), meaning it is less reactive than hydrogen and cannot reduce H^+ to H_2 . **Q39.** (i)



$2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$ **Q40.** (i) The high hydration enthalpy of V^{2+} compensates for the ionization energies, making the reduction of V^{3+} to V^{2+} favorable. (ii) High number of unpaired d-electrons leads to strong interatomic metallic bonding, requiring high energy to break. (iii) Electron-electron repulsions between the paired electrons in the nearly filled d-orbitals overcome the increased nuclear charge, causing a slight expansion.

Q41. (i) Cu^{2+} oxidises iodide to iodine: $2Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$. Therefore, CuI_2 does not exist. (ii) Increasing pH (adding base) converts orange dichromate ($Cr_2O_7^{2-}$) to yellow chromate (CrO_4^{2-}). (iii) Scandium (Sc) or Zinc (Zn). **Q42.** (i)



Q43. (A) Step 1: Fusion of MnO_2 with KOH and air to form potassium manganate:

$2MnO_2 + 4KOH + O_2 \xrightarrow{\Delta} 2K_2MnO_4 + 2H_2O$. Step 2: Electrolytic or acidic oxidation of manganate to permanganate: $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$. (B)

$Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$. (C) Similarity: Both show a prominent +3 oxidation state. Difference: Actinoids are all radioactive and show a much wider range of oxidation states than lanthanoids. **Q44.** (i) (A) is K_2MnO_4 (Potassium manganate), (B) is $KMnO_4$ (Potassium permanganate), (C) is KIO_3 (Potassium iodate), (D) is I_2 (Iodine). (ii) Alkaline:

$2MnO_4^- + H_2O + I^- \rightarrow 2MnO_2 + 2OH^- + IO_3^-$. Acidic:

$2\text{MnO}_4^- + 10\text{I}^- + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{I}_2 + 8\text{H}_2\text{O}$. **Q45.** (i) Ti(IV) has a d^0 noble gas configuration. (ii) Highly electronegative halogens (like F) stabilize high oxidation states via high lattice energy or high bond enthalpy. (iii) Zn ($3d^{10}4s^2$) has no unpaired d-electrons to participate in metallic bonding. (iv) Sc^{3+} is $3d^0$ (diamagnetic). Cu^{2+} is $3d^9$ (1 unpaired electron = paramagnetic). (v) $\text{Cr}^{2+} \rightarrow \text{Cr}^{3+}$ yields stable t_{2g}^3 . $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ yields stable d^5 . The t_{2g}^3 configuration is highly stable in water, making Cr^{2+} a stronger reducing agent. **Q46.** (A) 1. Fusion: $4\text{FeCr}_2\text{O}_4 + 8\text{Na}_2\text{CO}_3 + 7\text{O}_2 \rightarrow 8\text{Na}_2\text{CrO}_4 + 2\text{Fe}_2\text{O}_3 + 8\text{CO}_2$. 2. Acidification: $2\text{Na}_2\text{CrO}_4 + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{Cr}_2\text{O}_7 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$. 3. Displacement: $\text{Na}_2\text{Cr}_2\text{O}_7 + 2\text{KCl} \rightarrow \text{K}_2\text{Cr}_2\text{O}_7 + 2\text{NaCl}$. (B) Fe^{2+} (d^6) has 4 unpaired electrons: $\mu = \sqrt{4(6)} = 4.90$ BM. Mn^{2+} (d^5) has 5 unpaired electrons: $\mu = \sqrt{5(7)} = 5.92$ BM. Mn^{2+} is more paramagnetic. **Q47.** (i) The atomic radii of 4d and 5d transition series elements become almost identical (e.g., Zr and Hf). (ii) The 5f electrons provide even poorer shielding from the increasing nuclear charge than 4f electrons. (iii) Ce is $[\text{Xe}]4f^15d^16s^2$. Ce^{4+} is stable (noble gas config) but readily accepts an electron to return to the most stable lanthanoid +3 state, making it a good oxidising agent. **Q48.** (A) $\text{TiO} > \text{VO} > \text{CrO} > \text{FeO}$. The basic character decreases as the effective nuclear charge increases (and atomic size decreases) across the period, increasing covalent character. (B) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ has water acting as ligands causing crystal field splitting of the d^9 orbital, allowing a d-d transition (blue). Anhydrous CuSO_4 lacks ligands. ZnSO_4 is d^{10} (no empty d-orbital for transition). (C) Chromate ion, CrO_4^{2-} or Dichromate ion, $\text{Cr}_2\text{O}_7^{2-}$ (Cr is in Group 6, ox state is +6). **Q49.** (A) (1) They exhibit variable oxidation states, forming unstable intermediates. (2) They provide a large surface area for adsorption. Example: V_2O_5 in the Contact Process or Fe in Haber's Process. (B) (1) Small size and high nuclear charge of metal ions. (2) Availability of vacant d-orbitals to accept lone pairs from ligands. (C) It converts to yellow chromate. $\text{Cr}_2\text{O}_7^{2-} + 2\text{OH}^- \rightarrow 2\text{CrO}_4^{2-} + \text{H}_2\text{O}$. **Q50.** (i) Similar atomic sizes allow them to replace each other in a crystal lattice. (ii) The trapped non-metal atoms restrict the movement of metal atoms, increasing hardness, and occupy reactive sites, increasing inertness. (iii) Basic: MnO . Acidic: Mn_2O_7 . (iv) Scandium (Sc) shows only +3 oxidation state.

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